PA1 Write Up Report

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Solid: Square pyramid | Depth Offset: (+0.15, +0.15)

E1 — Depth Test vs. Draw Order

• Prediction: Without depth testing, triangles are rendered in the order they are drawn (painter's algorithm), causing later shapes/faces to overlap earlier ones regardless of 3D position. With depth testing enabled, pixel Z-depths are compared, ensuring correct spatial occlusion regardless of draw order.

• Empirical Result: Disabling depth test (Depth: OFF) causes faces that are drawn later to obscure closer faces. Toggling draw order (key S) makes the second shape render over the first. When depth testing is ON, spatial relationships remain correctly rendered regardless of draw order or toggling.

E2 — Near/Far Planes and Depth Testing

• Spatial Depth Mapping: In WebGL normalized device coordinates (NDC), clipped coordinates map Z from [-1, 1] to Depth Buffer [0.0, 1.0]. Z = -1.0 corresponds to the near plane (0.0 depth), and Z = 1.0 corresponds to the far plane (1.0 depth). Thus, lower Z values are closer to the camera.

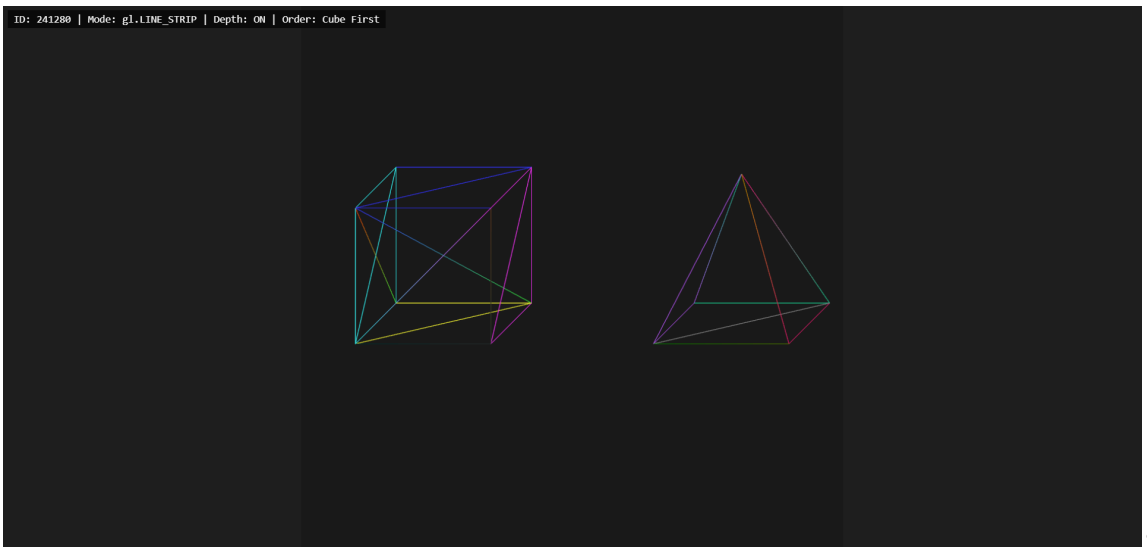
• (a) gl.depthFunc(gl.GREATER) with default gl.clearDepth(1.0): Nothing renders (screen remains blank). Incoming depth values from objects (less than 1.0) fail the GREATER test against the initial depth buffer value of 1.0.

• (b) gl.depthFunc(gl.GREATER) with gl.clearDepth(0.0): Geometry renders inverted in terms of depth priority, meaning further fragments overwrite closer ones if drawn later.

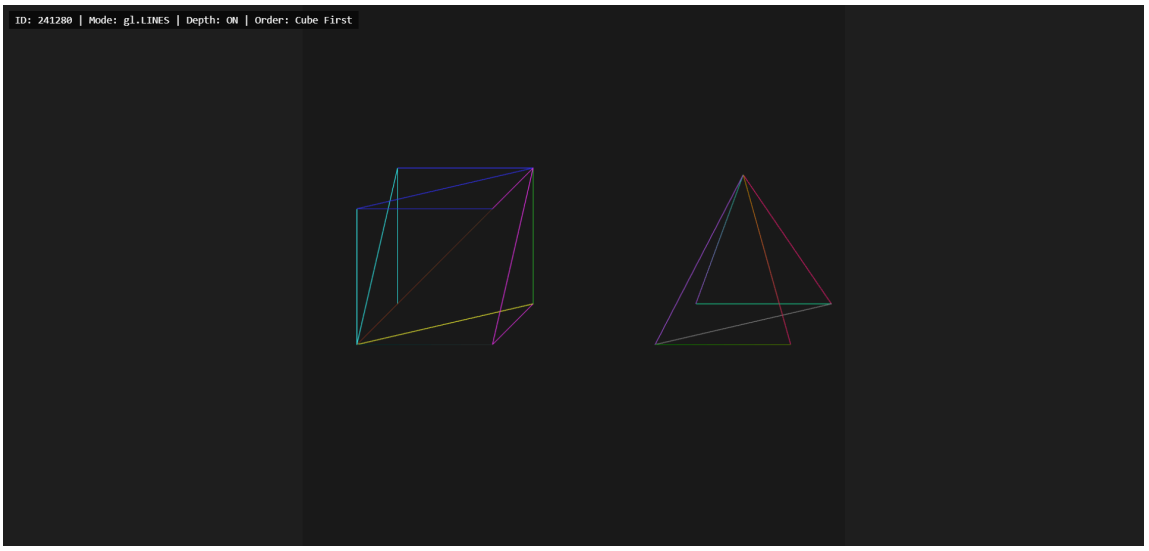
E3 — Primitive Rendering Modes



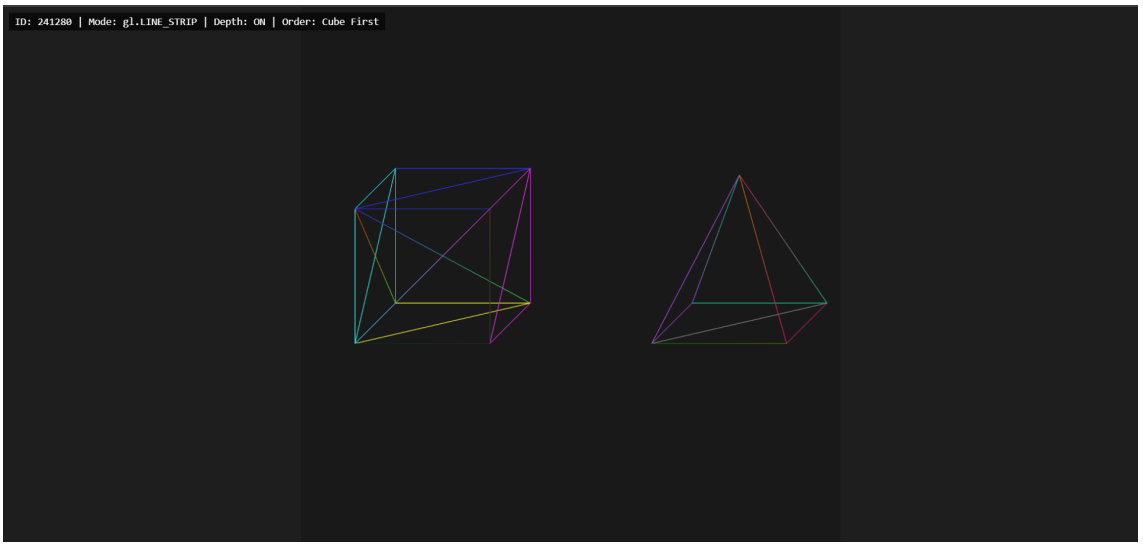
TRIANGLES: Interprets every 3 vertices as an independent triangle (correct solid mesh).



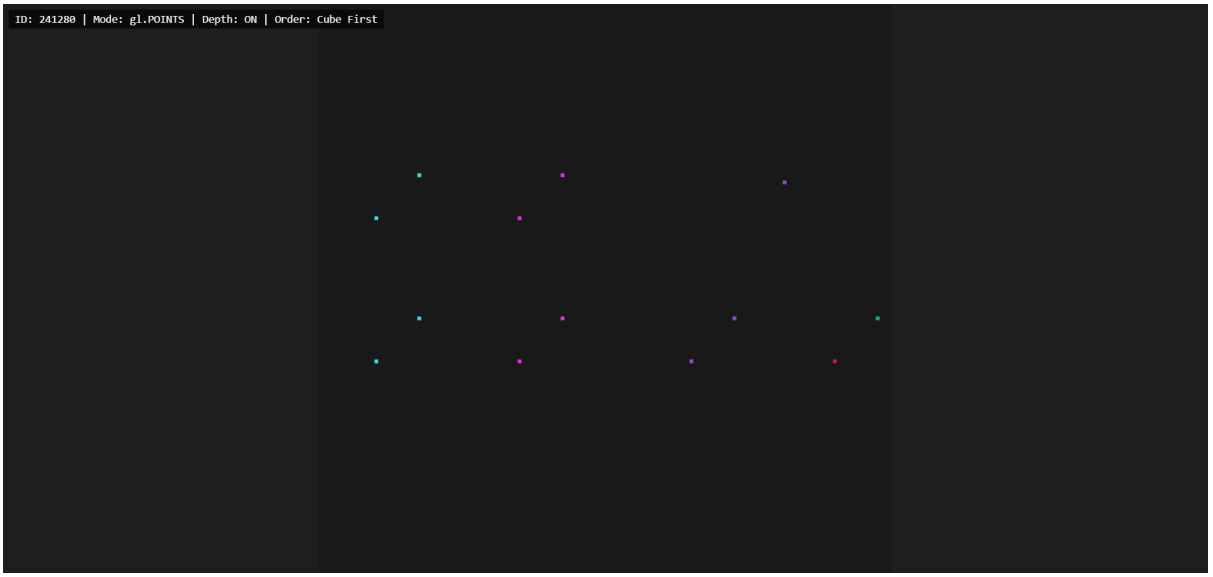
LINE_LOOP: Connects all vertices in sequence with a continuous line and closes the loop from last to first vertex, creating diagonal cross-lines across separate faces.



LINES: Connects every pair of 2 vertices into isolated line segments, leaving every third edge missing.



LINE_STRIP: Connects vertices sequentially with line segments without closing the final loop.



POINTS: Renders each vertex as an isolated point on the screen.

E4 — Vertex Attribute Array Mismatch
• Impact: Binding the wrong buffer (e.g., color buffer instead of position buffer) before gl.vertexAttribPointer causes WebGL to interpret color data as vertex coordinates. The geometry distorts or disappears. WebGL does not throw an error because the buffer size and data type (float32) match the expected input.

E5 — Clipping Behavior
• Point (0.0, 0.0, 1.2): Clipped by the Far Clipping Plane because its Z-coordinate (1.2) exceeds the maximum NDC range of +1.0.
• Point (1.4, 0.0, 0.0): Clipped by the Right Viewport Plane because its X-coordinate (1.4) exceeds the maximum NDC range of +1.0.

E6 — Buffer Memory Calculations
• Total Vertices = 36 (Cube: 6 faces * 2 triangles * 3 vertices) + 18 (Pyramid: 4 side faces * 3 vertices + 2 base triangles * 3 vertices) = 54 vertices.
• Position Buffer Size = 54 vertices * 3 components (x,y,z) * 4 bytes (Float32) = 648 bytes.
• Color Buffer Size = 54 vertices * 4 components (r,g,b,a) * 4 bytes (Float32) = 864 bytes.
• Verified via gl.getBufferParameter(gl.ARRAY_BUFFER, gl.BUFFER_SIZE).

Development Log
• Issue encountered: Initial pyramid faces had misaligned color assignments during primitive mode toggles.
• Solution: Fixed index offset logic in geometry arrays and re-bound buffers appropriately during render loop.
• Commit hash: b84a92c103f9e8a7d1c24e65f3a19028e45a1902